```
atmega32.upload.protocol=arduino
atmega32.upload.maximum_size=30720
atmega32.upload.speed=19200
atmega32.bootloader.low_fuses=0xFF
atmega32.bootloader.high_fuses=0xCA
atmega32.bootloader.path=atmega32
atmega32.bootloader.file=ATmega32BOOT.hex
atmega32.bootloader.unlock_bits=0x3F
atmega32.bootloader.lock_bits=0xCF
atmega32.build.mcu=atmega32
atmega32.build.f_cpu=16000000L
atmega32.build.core=arduino
atmega32.build.variant=mega32
```

Define the hardware of ATmega32. For this, search for directory location hardware\arduino\varients and create a new directory named mega32. Inside this new directory, add a new file named pins_arduino.h by copying it from Standard Directory.

This file includes digital pins, analogue pins, timer pins, pwm pins, ports and port bits. The numbers of digital pins, analogue pins and pwm pins are changed as below:

```
#define NUM_DIGITAL_PINS 32
#define NUM_ANALOG_INPUTS 8
#define analogInputToDigitalPin(p) ((p
< 8) ? (p) + 24 : -1)
#define digitalPinHasPWM(p) ((p) == 3 ||
(p) == 12 || (p) == 13 || (p) == 15)
```

Pin change interrupt (PCINT) is removed as it is not applicable for ATmega32. Constants are changed, that is, change for SS, MOSI, MISO, SCK, SDA and SCL as per ATmega32 hardware pin out.

```
static const uint8_t SS = 4;
static const uint8_t MOSI = 5;
static const uint8_t MISO = 6;
static const uint8_t SCK = 7;

static const uint8_t SDA = 17;
static const uint8_t SCL = 16;
```

Constants for analogue pins are defined as follows:

```
static const uint8_t A0 = 31;
static const uint8_t A1 = 30;
static const uint8_t A2 = 29;
static const uint8_t A3 = 28;
static const uint8_t A4 = 27;
static const uint8_t A5 = 26;
static const uint8_t A6 = 25;
static const uint8_t A7 = 24;
```

Remaining modifications can be made as given below:

```
const uint16_t PROGMEM port_to_mode
_PGM[] = {
        NOT_A_PORT,
        (uint16_t) &DDRA,
        (uint16_t) &DDRB,
        (uint16_t) &DDRC,
        (uint16_t) &DDRD,
};

const uint16_t PROGMEM port_to_
output_PGM[] = {
        NOT_A_PORT,
        (uint16_t) &PORTA,
        (uint16_t) &PORTB,
        (uint16_t) &PORTC,
        (uint16_t) &PORTD,
};

const uint16_t PROGMEM port_to_input
_PGM[] = {
        NOT_A_PORT,
        (uint16_t) &PINA,
        (uint16_t) &PINB,
        (uint16_t) &PINC,
        (uint16_t) &PIND,
};

const uint8_t PROGMEM digital_pin_to_
port_PGM[] = {
        PB, /* B *0* */
        PB,
        PB,
        PB,
        PB,
        PB,
        PB,
        PB,
        PD, /* D *8* */
        PD,
        PD,
        PD,
        PD,
        PD,
        PD,
        PC, /* C *16* */
        PC,
        PC,
        PC,
        PC,
        PC,
        PC,
        PA, /* A *24* */
        PA,
        PA,
        PA,
        PA,
        PA,
        PA,
        PA,
};
const uint8_t PROGMEM digital_pin_to_
bit_mask_PGM[] = {
        _BV(0), /* digital pin 0, port
                        B */
        _BV(1), //digital pin 1
        _BV(2), //digital pin 2
        _BV(3), //digital pin 3
        _BV(4), //digital pin 4
        _BV(5), //digital pin 5
        _BV(6), //digital pin 6
        _BV(7), //digital pin 7
        _BV(0), /* digital pin 8, port
                        D */
        _BV(1), //digital pin 9
        _BV(2), //digital pin 10
        _BV(3), //digital pin 11
        _BV(4), //digital pin 12
        _BV(5), //digital pin 13
        _BV(6), //digital pin 14
        _BV(7), //digital pin 15
        _BV(0), /*digital pin 16, port
                        C */
        _BV(1), //digital pin 17
        _BV(2), //digital pin 18
        _BV(3), //digital pin 19
        _BV(4), //digital pin 20
        _BV(5), //digital pin 21
        _BV(6), //digital pin 22
        _BV(7), //digital pin 23
        _BV(7), /* digital pin 24,
      port A, note descending order */
        _BV(6), //digital pin 25
        _BV(5), //digital pin 26
        _BV(4), //digital pin 27
        _BV(3), //digital pin 28
        _BV(2), //digital pin 29
        _BV(1), //digital pin 30
        _BV(0), //digital pin 31
};

const uint8_t PROGMEM digital_pin_to_
timer_PGM[] = {
        NOT_ON_TIMER, /* 0 - port B */
```